

Factoring Algebraic Expressions (Part III) Homework

Factor.

① $x^4 + 14x^3 + 45$

② $y^4 + 13y^2 + 12$

③ $b^4 + 10b^2 + 21$

④ $x^4 + 13x^2 + 22$

⑤ $y^4 - 9y^2 - 22$

⑥ $b^4 - 7b^2 + 10$

⑦ $x^4 + 2x^2 - 3$

⑧ $y^4 + 2y^2 - 24$

⑨ $2b^4 + 5b^2 - 12$

⑩ $3x^4 - 7x^2 - 20$

⑪ $12y^4 - 52y^2 + 35$

⑫ $4x^3 - 12x^2 + 3x - 9$

⑬ $7y^4 - 21y^3 + 2y - 6$

⑭ $b^3 - 2b^2 - 9b + 18$

⑮ $2x^3 + 2x^2 - 5x - 5$

⑯ $y^3 + y^2 - 16y - 16$

⑰ $b^3 - 4b^2 - b + 4$

⑱ $12x^3 + 14x^2 + 6x + 7$

⑲ $5y^3 - 40y^2 - 4y + 32$

Answers

- ① $(x^2+9)(x^2+5)$
- ② $(y^2+12)(y^2+1)$
- ③ $(b^2+3)(b^2+7)$
- ④ $(x^2+11)(x^2+2)$
- ⑤ $(y^2-11)(y^2+2)$
- ⑥ $(b^2-2)(b^2-5)$
- ⑦ $(x+1)(x-1)(x^2+3)$
- ⑧ $(y+2)(y-2)(y^2+6)$
- ⑨ $(2b^2-3)(b^2+4)$
- ⑩ $(3x^2+5)(x+2)(x-2)$
- ⑪ $(2y^2-7)(6y^2-5)$
- ⑫ $(4x^2+3)(x-3)$
- ⑬ $(7y^3+2)(y-3)$
- ⑭ $(b-2)(b+3)(b-3)$
- ⑮ $(2x^2-5)(x+1)$
- ⑯ $(y+4)(y-4)(y+1)$
- ⑰ $(b+1)(b-1)(b-4)$

$$\textcircled{18} (2x^2+1)(6x+7)$$

$$\textcircled{19} (5y^2-4)(y-8)$$